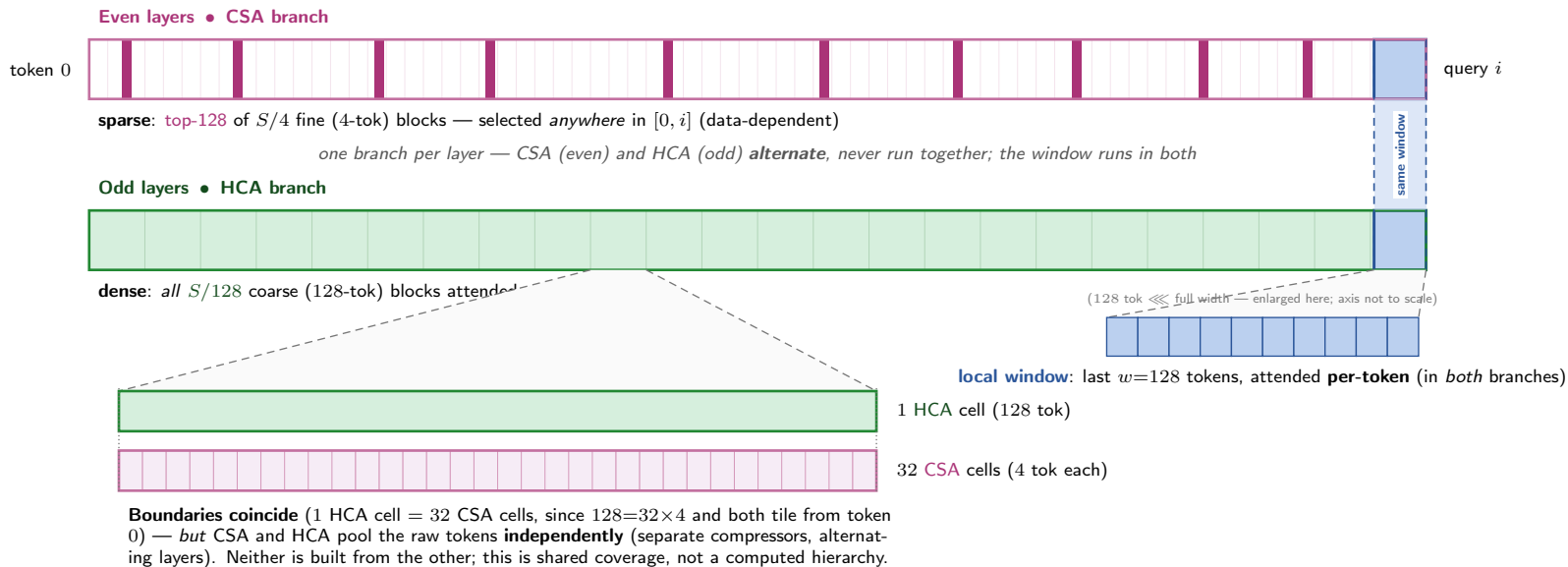


MORPH Attention — Context Coverage at Scale

two parallel branches over one causal context — CSA (sparse) and HCA (dense) on alternating layers, plus the local window



 CSA: sparse selected fine block

 HCA: dense coarse block

 window: per-token local (both branches)

At $S=1\text{M}$: HCA attends all $S/128 = 8,192$ coarse blocks (dense); CSA keeps top-128 of $S/4 = 262,144$ fine blocks (sparse, $\approx 0.05\%$); window = 128 tokens. Block sizes symbolic in S ($n_{\text{blk}}=S/m$, computed live).

Invariants: CSA and HCA *alternate by layer* (even $\rightarrow$ CSA, odd $\rightarrow$ HCA) — the two lanes never co-occur; the combined picture is what the *stack* sees. The *window* is always on. All attention is causal $[0, i]$.